2026

# IRoC-U

ISRO Robotics Challenge - URSC

*Let's build a space robot*

**Proposal Report**

For IRoC-U 2026

by

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# 1. Description of ASCEND

ASCEND requires an autonomous micro-UAV to operate in a Mars-like, GPS-denied environment and perform terrain exploration, target detection, and precision return-to-base without human intervention. The UAV must rely entirely on onboard sensing, state estimation, and autonomous control to complete the mission.

Fig 1.1: Front View

Fig 1.2: Top View

Fig 1.3: Isometric View

Fig 1.4: Side View

Team Challengers proposes a fully autonomous quadcopter–docking system designed specifically for GNSS-, marker-, and external-aid-denied operation. The system consists of an X-configuration quadcopter optimized for precision flight and an autonomous docking station that enables post-mission data transfer and battery recharging

System Overview

- X-configuration quadcopter capable of stable hover and low-speed maneuvering
- Fully onboard autonomy for takeoff, exploration, target detection, return, and landing
- Vision- and inertial-based navigation without reliance on GPS or external infrastructure
- Autonomous docking station for alignment, charging, and data exchange

Platform Selection and Feasibility

The proposed UAV platform was evaluated against both commercial and academic quadcopter systems to assess feasibility, cost-efficiency, and mission suitability. Unlike high-end commercial platforms such as the DJI Matrice 100 or Parrot Anafi USA, the proposed system emphasizes modularity, affordability, and full autonomy, making it better suited for experimental development and competition constraints.

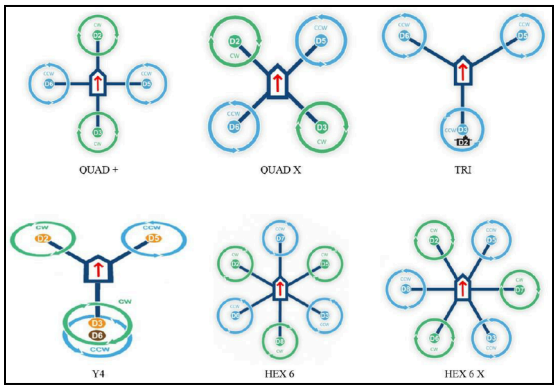


Fig 1.5

A quadcopter configuration was selected after evaluating multiple aerial platforms:

- Fixed-wing aircraft were unsuitable due to their inability to hover and perform precision landings
- Larger multirotor configurations increase redundancy but significantly raise mass, power consumption, and system complexity
- The quadcopter provides the optimal balance between payload capacity, control accuracy, energy efficiency, and mechanical simplicity

**Payload and Design Constraints**

The quadcopter is capable of carrying a payload of up to 500 g, sufficient for the onboard flight computer, vision sensors, and supporting electronics with adequate design margin. The complete system operates within the 2 kg mass limit and is constructed using off-the-shelf components, including high-torque BLDC motors, a carbon-fiber airframe, Li-Po battery, and a Pixhawk Cube Orange flight controller. This approach balances durability, payload capability, and system reliability while simplifying integration and maintenance

**Docking Concept**

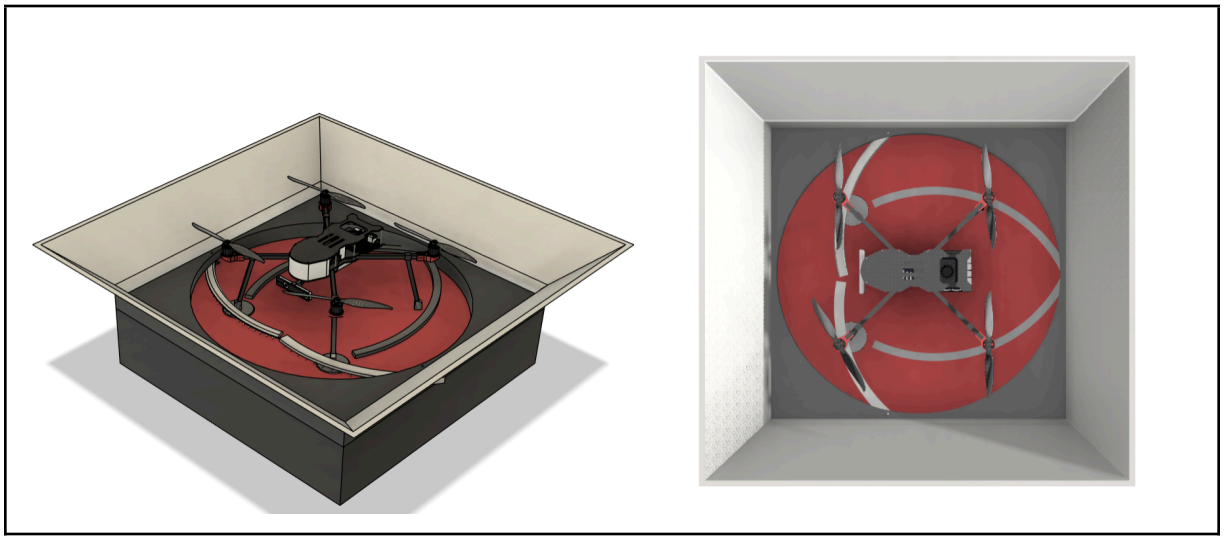

Fig 1.6

The docking station is designed to perform accurate, reliable, and fully autonomous landing and alignment under GPS-, GNSS-, LPS-, and external-marker-denied conditions. It features a single circular base platform that provides a symmetrical and tolerant landing surface. Embedded circular metal contact plates serve as electrical charging interfaces, while the circular geometry reduces sensitivity to yaw misalignment during landing. The docking system enables safe alignment, power transfer, and data exchange without human intervention.

Overall, the proposed ASCEND system prioritizes robust autonomy, precision control, and competition compliance, ensuring reliable mission execution in a fully GNSS-denied environment.

# 2. System Architecture

The system architecture of ASCEND combines onboard sensing, high-level autonomy, and low-level flight control to enable reliable autonomous operation without external navigation aids. The architecture emphasizes modularity, real-time performance, and seamless sensor–controller integration.

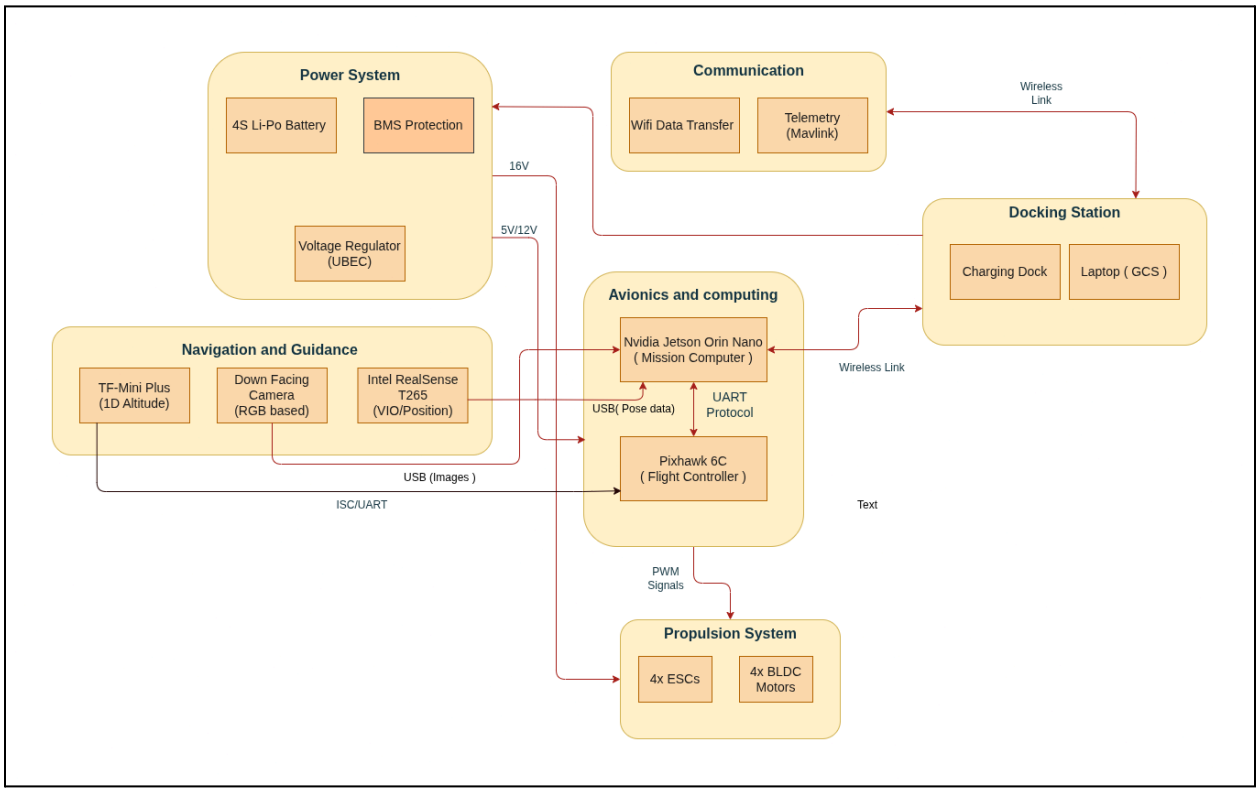


Figure 2.1

The hardware design is centered around a dual-computer avionics stack, utilizing an Nvidia Jetson Orin Nano as the primary mission computer and a Pixhawk 6C for real-time flight control. Power management is handled by a 4S Li-Po battery system equipped with BMS protection, which distributes raw voltage to the propulsion system's ESCs and motors while routing regulated 5V and 12V lines to the sensitive electronics. The navigation suite integrates multiple sensor modalities, including a TF-Mini Plus for 1D altitude precision, a down-facing RGB camera for visual data, and the Intel T265 for VIO-based positioning, all interfacing with the mission computer via USB or UART. Communication is maintained through dual links: a telemetry module handles MAVLink traffic for flight data, while a high-bandwidth WiFi link transfers data between the onboard computer and the docking station's Ground Control Station (GCS).

**Software Architecture (ROS 2 Implementation)**

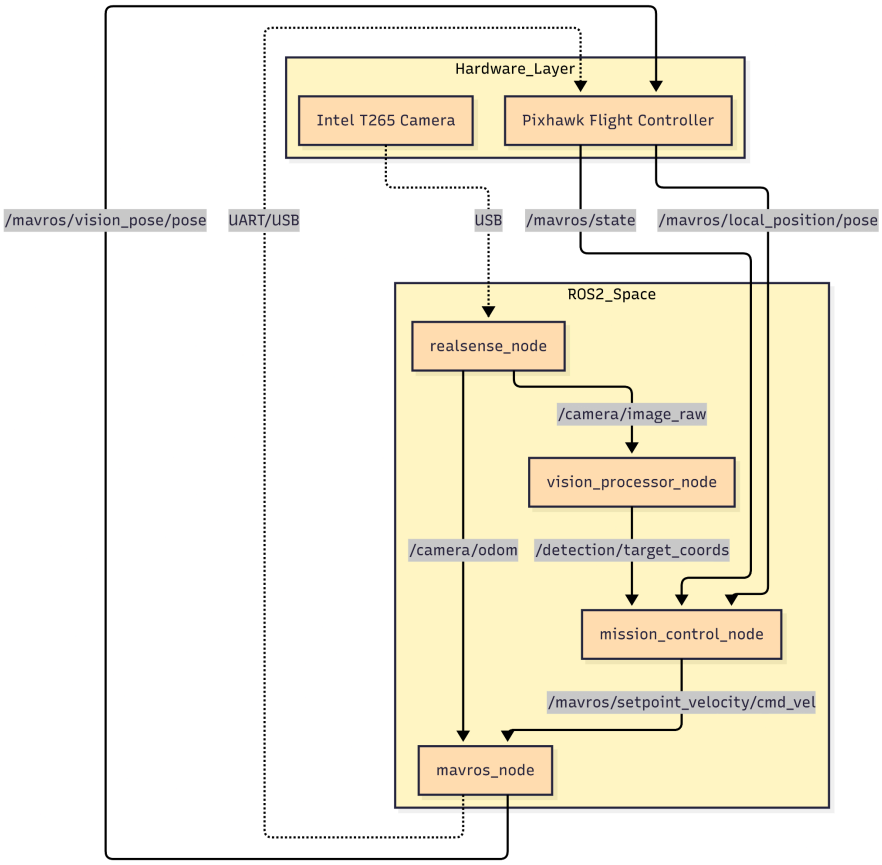


Figure 2.2

The software stack is built on a ROS 2 framework designed to decouple low-level flight stabilization from high-level perception and control logic. The architecture bridges the hardware layer—consisting of the Intel T265 and Pixhawk flight controller—with the application layer through a series of modular nodes. The realsense_node acts as the primary driver, ingesting raw data via USB and splitting it into two parallel streams: visual odometry (/camera/odom) for state estimation and raw video frames (/camera/image_raw) for the perception pipeline. While the mavros_node handles the critical MAVLink bridge to keep the Pixhawk updated with position data, the vision_processor_node executes object detection algorithms to output target coordinates. The central decision-making unit, the mission_control_node, subscribes to both the target coordinates and the drone's local position to generate real-time velocity commands (cmd_vel), closing the loop by sending these setpoints back to the flight controller for execution.

# 3. Identification of components with their specifications

3.1 Components Rotarcraft:

1. Mechanical Components

Frame: The ASCEND rotarcraft drone is built around a 500mm carbon fiber quadcopter frame, which forms the structural backbone of the vehicle.This custom quadcopter frame is fabricated from high-quality carbon fiber, making it lightweight, strong, and highly rigid. The carbon-fiber arms provide excellent structural integrity with minimal weight, ensuring improved flight efficiency and reduced vibrations. The optimized arm thickness and reinforced motor mounts enhance durability and minimize damage during hard landings. The rigid frame design improves overall stability, enabling precise control, faster forward flight, and better payload handling compared to conventional frames.

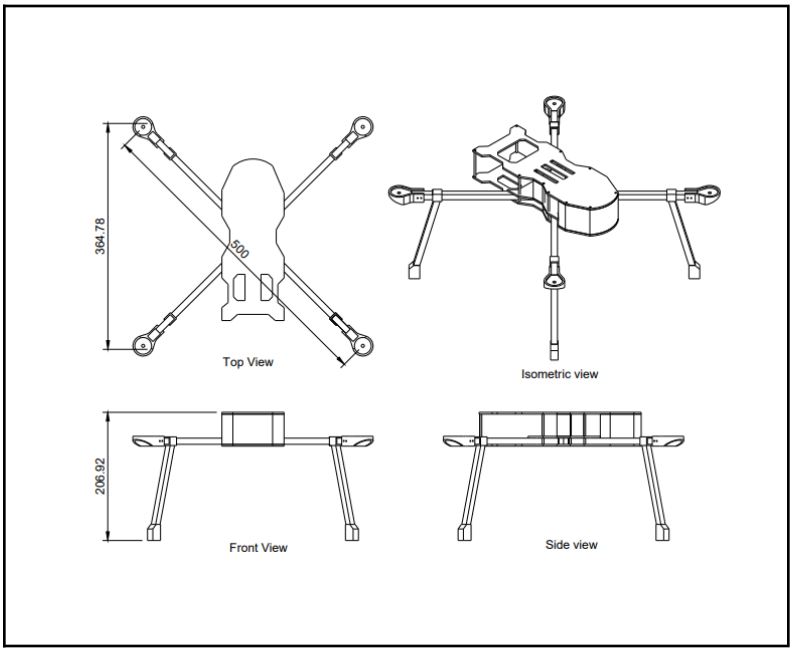


Fig 3.1.1

The drone's propulsion system consists of four AIR2216II KV920 brushless motors, each paired with an AIR 20A electronic speed controller (ESC). These motors provide sufficient thrust to lift the drone and its onboard electronics while maintaining precise control for maneuvering. The T1045 propellers (10×4.5) are arranged in a CW/CCW configuration to convert motor torque into lift effectively. While carbon fiber propellers were considered for slightly

higher efficiency, nylon-glass propellers were chosen for their robustness and ability to withstand minor impacts and vibrations, which is important for repeated autonomous sortie

2. Electrical Components

(a) Power Distribution

A 4S 3300 mAh Li-Po battery powers the entire drone, supplying all motors and ESCs. A separate 5V BEC (2–3A) provides clean, regulated power for the Jetson Orin Nano, the Cube Orange+ flight controller, and sensors. This setup ensures stable voltage and protects electronics from motor noise.

(b) Propulsion System

The drone uses four AIR2216II KV920 motors, drawing 8–12A in hover and up to 22A at full load.
Each motor is controlled by a 20A ESC, capable of short 30A bursts.
T1045 propellers generate the necessary lift and influence efficiency, motor load, and stability.

(c) Computation Hardware

The Cube Orange+ flight controller handles stabilization and uses about 250–300 mA at 5 V.
The Jetson Orin Nano runs high-level processing and AI tasks, drawing 2–3 A at 5 V depending on workload.

3. Sensors.

(a) Intel RealSense T265 – Visual-Inertial Odometry (VIO) (Fig 3.1.2)
Used for GPS-denied navigation. It estimates the drone's position and orientation using stereo vision + internal inertial sensors. Helps the drone know where it is in indoor/complex environments.
Position update:
$P(t)=P(t-1)+v(t)\Delta t+21a(t)\Delta t2$
Velocity update:
$v(t) = v(t-1) + a(t)*\Delta t$

(b) TFmini-S 1D LiDAR – Altitude Measurement (Fig-3.1.3)
Provides very accurate altitude at low heights (0–5 m), where barometers are unreliable. Essential for stable hovering and smooth autonomous landing.
Time-of-flight distance:
$d = (c * t) / 2$

where

d = distance

c = speed of light

t = round-trip travel time

(c) Raspberry Pi Camera Module 3 – Vision & Detection (Fig 3.1.4)

Used for object detection, feature tracking, and target identification. Works with Jetson's computer vision models to locate the docking station or mission targets.

Harris corner response:

$$R = det(M) - k(trace(M)^2$$

where

M = structure tensor

k ≈ 0.04

Appendix: Figures

| | | |
|---|---|---|
| Fig 3.1.2:Intel RealSense T265 | Fig 3.1.3: TFmini-S 1D LiDAR | Fig 3.1.4: Raspberry Pi Camera Module 3 |

**Algorithms and Software**

The software stack integrates low-level control, high-level perception, and autonomous mission management. Low-level flight stabilization is handled by the PX4 autopilot firmware on the flight controller, which manages motor commands and ensures stable flight. High-level autonomy runs on the Jetson Orin Nano using ROS 2, with custom nodes in C++ and Python handling SLAM, path planning, and feature detection.

Sensor data from the IMU, LiDAR, and T265 camera are fused using an Extended Kalman Filter (EKF) to generate an accurate estimate of the drone's position and orientation. Path planning algorithms, including A* and RRT, allow the drone to explore a designated area efficiently while avoiding obstacles. The software also handles autonomous target detection, coordinate logging, and image capture for validation at the docking station. For returning and landing, the LiDAR measures height, and the T265 camera assists with precise alignment, ensuring the drone docks safely. Overall, the algorithms are designed to allow the drone to operate fully autonomously, performing multiple sorties without human intervention.

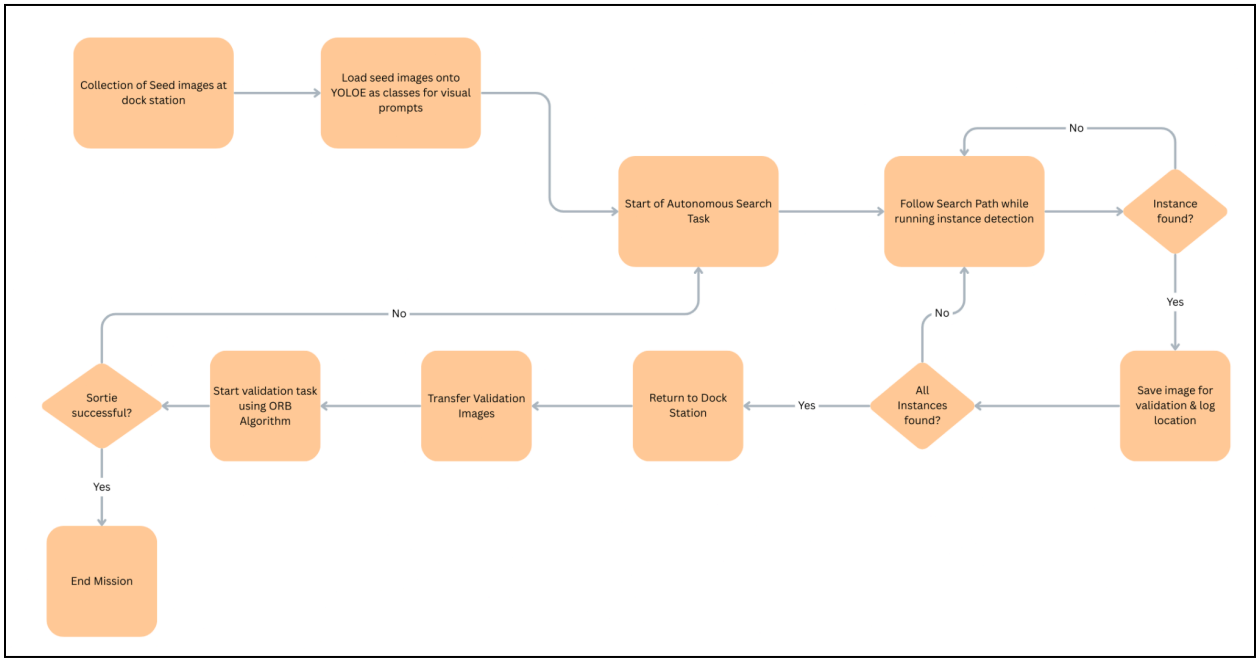


Fig 3.1.5

The mission workflow begins with the Seeding Task, where 3–5 sample feature images are collected at the dock station and loaded onto the YOLO-E model as visual prompts to define target classes. Once the Autonomous Search Task starts, the UAV follows a pre-defined path while running real-time instance detection to locate targets that are similar to the seed images. When an instance is found, the system saves a verification image and logs the local estimated coordinates; if not all instances are found, the UAV continues its search. After identifying the required targets, the UAV executes the Return Task to autonomously land at the dock station. The captured data is then transferred to the base station for the Validation Task, where an ORB pattern-matching algorithm compares the findings against the original seeds. If the sortie is successful, the mission ends; otherwise, the UAV may initiate repeat sorties to complete the objectives.

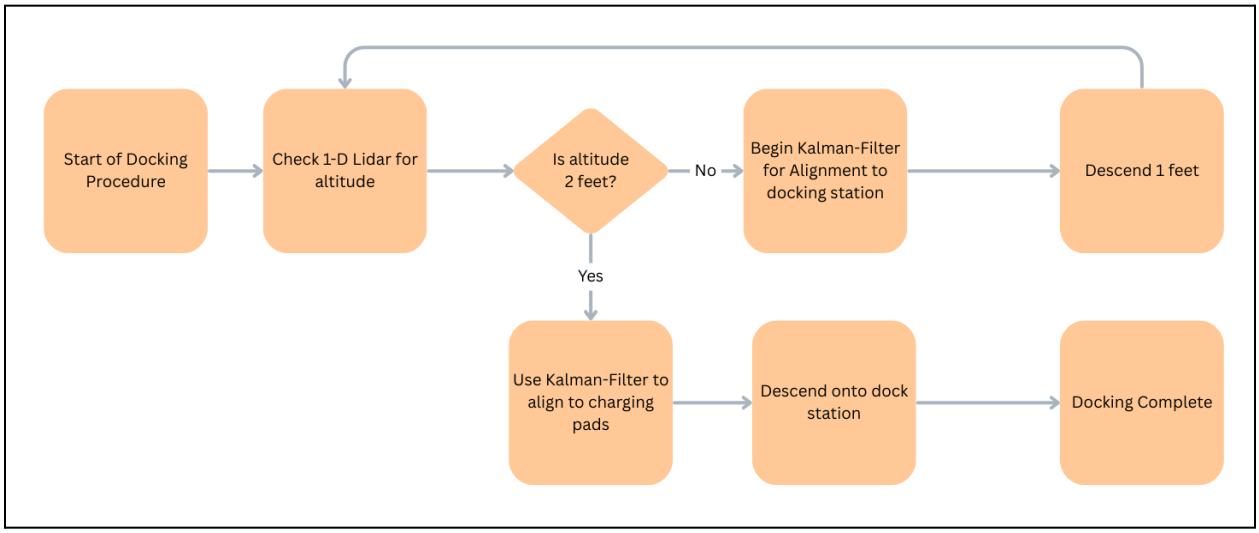


Fig 3.1.6

The process begins with the UAV utilizing a 1-D Lidar to verify its altitude, ensuring it adheres to the strict prohibition against using GNSS or external navigation aids. If the altitude is not at the 2-foot threshold, the system employs a Kalman-Filter for alignment to the docking station while descending in increments; once the 2-foot altitude is reached, the filter specifically aligns the UAV with the charging pads to ensure a precise landing. This autonomous descent onto the dock station is critical because manual alignment or human intervention is strictly forbidden. Upon completion of the docking, the UAV can then fulfill the mandatory requirements of transferring image data and coordinates to the base station while simultaneously recharging its power system through the designated interfaces.

**3.2 Components Dockstation:**

The docking station is designed to enable safe autonomous landing, secure positioning, charging, and data transfer of the drone under GPS- and marker-denied conditions. It supports reliable interaction between the aerial platform and ground system by combining simple mechanical actions with controlled electrical and communication interfaces.

Cleaning and Locking Mechanism

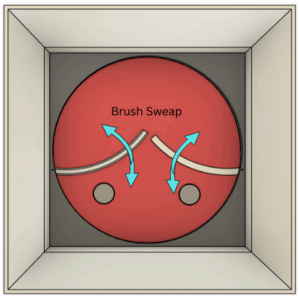


Fig 3.2.1: Cleaning

Fig 3.2.2: Aligning

Before landing, the docking station cleans the drone's charging contact surfaces to ensure a reliable electrical connection. The ESP32 controller receives a landing-approach command from the docking station computer, after which the cleaning mechanism is activated. After landing, a locking and alignment mechanism holds the drone firmly in place and corrects minor positional or orientation misalignment.

Charging System

Once proper alignment is achieved, the charging system establishes electrical contact and safely charges the drone's battery while it remains docked. The drone's landing legs are fitted with metallic contact pads, which interface with the docking station's charging contacts to ensure a reliable and stable electrical connection during charging.

- Battery energy capacity: 48.84 Wh (14.8 V, 3.3 Ah)
- Charging current: 6.0 A
- Contact resistance: 0.025 Ω
- Voltage drop at charging plates: 0.15 V
- Power consumption: ~112 W
- Total charging time: ~39 minutes

Data Transfer

After docking, the drone connects to the docking station via wireless communication, enabling reliable data transfer between the aerial platform and the ground system.The data transfer system uses a combination of Wi-Fi and Bluetooth communication to ensure reliable, secure, and repeatable data exchange between the drone and the docking station after successful docking.

# 4. ASCEND Realization Plan

| No. | Hardware Details | Procurement Source (Market/Fabrication/3D Printing/…..) | Specifications/ Realization Plan | Quantity |
|---|---|---|---|---|
| 1 | Frame | Fabrication (Carbon Fiber PLA) | 450mm quadcopter frame, lightweight and rigid, includes mounting points for landing gear and electronics | 1 |
| 2 | Motors | Market | AIR2216II KV920 brushless, 4S LiPo compatible, 64g each, provides sufficient thrust for UAV | 4 |
| 3 | ESCs | Market | AIR 20A ESC, continuous current 20A, peak 30A, 53x22x7 mm, connects to motors via 3.5mm banana plugs | 4 |
| 4 | Propellers | Market | T1045 (10×4.5), nylon + glass fiber, 68g total for 4, CW/CCW setup | 4 |
| 5 | Battery | Market | 4S 3300mAh LiPo, 16.8V full charge, provides power to motors, flight controller, and Jetson | 1 |
| 6 | Flight Controller | Market | Cube Orange flight controller module | 1 |
| 7 | Companion Computer | Market | Jetson Orin Nano, GPU accelerated, runs ROS2 for SLAM, navigation, and mission control | 1 |
| 8 | Cooling Fan | Market | Fan for Jetson Orin Nano, maintains operating temperature during high-load computation | 1 |
| 9 | Visual Sensor | Market | Intel RealSense T265 camera, provides visual-inertial odometry (VIO) for GPS-denied navigation | 1 |
| 10 | 1D Lidar | Market | TFmini-S, for altitude measurement and safe landing | 1 |
| 11 | High-Current Switch | Market | XT60, 15–20A, for safe power on/off | 1 |

| | | | | |
|---|---|---|---|---|
| 12 | Power Regulation / Power Electronics Component | Market | Holybro UBEC 12A: 3–14S input, selectable 6/7.2/8/9.2 V output, 12 A continuous (24 A peak), high-efficiency UBEC with full protection features. | 1 |
| 13 | Landing Gear with Metal contacts | Fabrication | Lightweight landing assembly to support charging after landing in the Docking station | 1 |

# 1. Testing Plan

To ensure that the ASCEND system meets the requirements of autonomous navigation, perception, and mission execution without external navigation aids, a structured testing approach is adopted. Validation is carried out at subsystem and integrated system levels using high-fidelity PX4 SITL, ROS 2, and Gazebo simulations to verify sensing, autonomy, and safety prior to hardware deployment.

Subsystem-Level Testing

Camera (Downward-Facing RGB Camera)

1. Accuracy of visual feature detection
2. Consistency of image capture for documentation
3. Robustness under varying lighting and viewpoints

IMU (Inertial Measurement Unit)

1. Stability under static and dynamic conditions
2. Accuracy of acceleration and angular rate measurements

Visual-Inertial Odometry (VIO)

1. Position and velocity estimation accuracy
2. Orientation tracking in dynamic flight conditions

Mapping and Perception Module

1. Flat surface detection for landing
2. Obstacle detection and avoidance
3. Consistency of real-time map generation

Flight Controller and Navigation

1. Autonomous takeoff and landing
2. GPS-denied navigation stability
3. Autonomous return-to-base execution

Integrated System Testing

1. Coordination between sensing, estimation, and control subsystems
2. End-to-end autonomous mission execution
3. Robustness under environmental disturbances (lighting changes, sensor noise)

**Test Plan Summary Table**

| Test No. | Test Name | Description | Procedure | Expected Outcome |
|---|---|---|---|---|
| 1 | Visual Feature Detection | Verify camera's ability to detect target features | Simulate seeded features in Gazebo arena and evaluate detection confidence | Correct detection in >95% of trials |
| 2 | Image Capture Validation | Verify quality of captured verification images | Capture images at different altitudes and viewpoints | Images suitable for feature validation |
| 3 | IMU Stability | Test IMU behavior in static and dynamic flight | Log IMU data during hover and maneuvers | Stable acceleration and rate estimates |
| 4 | VIO Position Tracking | Validate position and velocity estimation | Compare VIO output with commanded motion | Position error within acceptable bounds |
| 5 | Autonomous Takeoff | Verify safe autonomous takeoff | Execute takeoff sequence in simulation | Smooth ascent and stable hover |
| 6 | Autonomous Landing | Validate safe landing at base station | Perform repeated landing trials | Landing within tolerance region |
| 7 | End-to-End Mission | Verify full ASCEND mission | Execute takeoff → search → return → land | Mission completed without intervention |
| 8 | Resilience Testing | Test behavior under faults | Simulate sensor dropouts and low energy | Safe recovery and controlled landing |

## 6. Brief of system specification

| Sl. No. | Description | Specification |
|---|---|---|
| 1. | Overall mass | 1689.24 |
| 2. | Overall dimensions (mm) | 500*500*206 |
| 3. | Power requirements(w) | 243.16 |
| 4. | Flight time per charge | 9.64 min |
| 5. | Number of propellers and their size | 4(10x45) |
| 6. | Features | |
| 7. | Others | If any |

## 7. Overview of the emergency response system.

| Emergency situation | Description | Solution |
|---|---|---|
| Power failure | An unexpected mid-flight power loss may cause the drone to crash, risking damage to both the drone and nearby areas. | Implement a battery health monitoring system and use an auxiliary battery to initiate an emergency landing before complete power loss. |
| Communication loss | i) A disruption in communication between the drone and the ground station can result in a potential loss of control.<br><br>ii) Loss of communication between the Pixhawk and Jetson can disrupt autonomous flight operations. | i) Enable an autonomous return-to-home (RTH) function. Use pre-programmed safe zones as fallback destinations.<br><br>ii) This issue is mitigated by utilizing PX4's offboard mode to maintain autonomous control during communication interruptions. |
| Obstacle collision | Inability to detect or avoid obstacles during flight may result in damage to the drone or surrounding objects. | Deploy real-time obstacle detection and avoidance algorithms using RANSAC-based point cloud analysis for precise avoidance. |

| | | |
|---|---|---|
| Landing zone obstruction | Insufficient precision during landing can lead to improper alignment with the dock, causing obstruction or unsafe landing conditions. | Landing accuracy issues are mitigated using vision-assisted precision landing |
| Environmental factors | Adverse conditions like water-ice, dry-ice and global dust stroms affect drone stability and camera vision. | Upon detection of camera degradation due to dust or environmental factors, the system activates an autonomous RTH sequence to restore sensor reliability. |
| System malfunctions | Failures in key components like theIMU, camera, or point cloud mapping, leading to operational errors. | Failures in essential components are managed through continuous health monitoring and fail-safe mechanisms that trigger autonomous RTH or controlled landing. |

# 8 Project management

To ensure the successful execution of the ASCEND mission, the project is managed by assigning clear roles and timelines. The following management strategy has been established.

8.1 Team Roles and Responsibilities

The team is divided into functional roles, with each team member having a task assigned to them.

| No. | Task | Main Responsibility | Deadline for Completion | Secondary Responsibility |
|---|---|---|---|---|
| 1 | Frame Assembly & Wiring | Poojith | | Tushar |
| 2 | Hardware and software Integration | Tanmay | | Ranjith |

| | | | | |
|---|---|---|---|---|
| 3 | Simulation and Testing | Sanjana | | Poojith |
| 4 | Docking station | Saalim | | Tushar |
| 5 | ROS Implementation | Ranjith | | Tanmay |
| 6 | Navigation and Mapping | Dhanushree | | Neil |
| 7 | Detection and Validation | Laksh | | Aman |
| 8 | Documentation and Report | Aman | | Neil |

**8.2 Project Schedule & Milestones**

The ASCEND project follows a milestone-driven schedule to ensure timely completion of design, development, and validation activities in alignment with the IRoC-U 2026 competition timeline. A combined hardware–software Gantt-based planning approach is adopted to track progress and manage dependencies between subsystems.

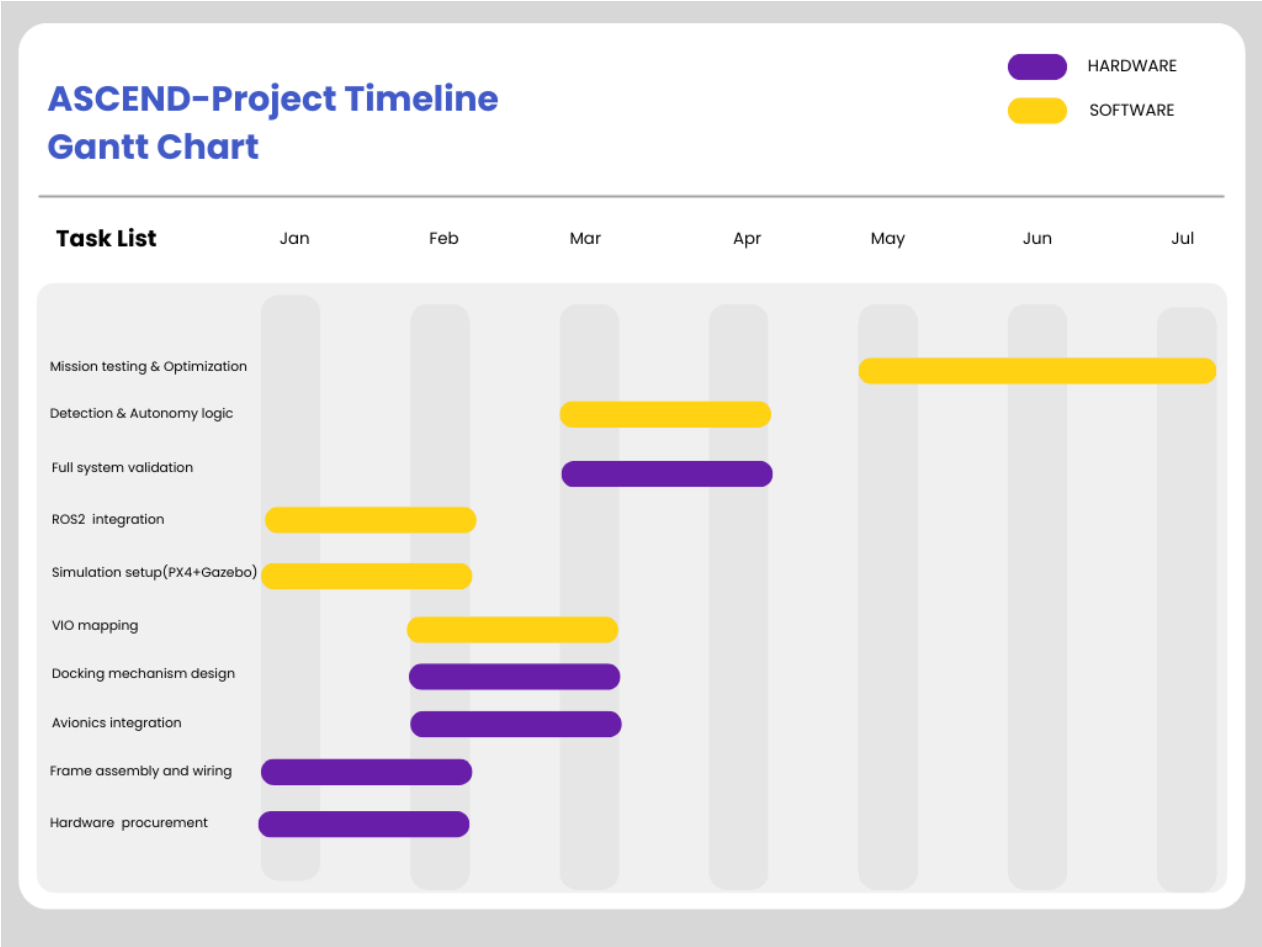


Fig 8.1

The development is structured into four major phases:

- Phase I – Concept Definition and Simulation Setup (Dec 2025 – Jan 2026):
  Finalization of system architecture and mission workflow. High-fidelity simulation is established using PX4 SITL, ROS 2, and Gazebo to validate autonomy, perception, and navigation algorithms in a GPS-denied environment.

- Phase II – Hardware Integration and Software Development (Jan 2026 – Mar 2026):
  Airframe assembly, avionics integration, and docking station development are carried out in parallel with the implementation of visual-inertial odometry, mapping, and feature detection algorithms.

- Phase III – Integrated Testing and Validation (Mar 2026 – Apr 2026):
  Subsystem and system-level testing is performed to validate autonomous takeoff, search, return, landing,

data transfer, and charging operations under competition-like conditions.

- Phase IV – Refinement and Competition Readiness (Apr 2026 – Jul 2026):
  Performance optimization, robustness testing, and iterative improvements are conducted based on evaluation feedback, leading to final system readiness for the field demonstration.

The internal milestones are synchronized with the official IRoC-U 2026 schedule to ensure progressive system maturity at each evaluation stage, culminating in a fully autonomous and competition-ready ASCEND system.

# 9. Novelty in the overall proposal

## 1. Optimized Drone Frame Design for Docking Compatibility

The proposed quadrotor frame is uniquely designed with a compact, symmetric cross-arm configuration as shown in Figures (top and bottom views). The arm geometry ensures uniform thrust distribution and maintains a low center of gravity, improving flight stability during autonomous descent and docking maneuvers. Unlike conventional drone frames, the landing structure is specifically aligned with the circular docking interface, allowing precise self-alignment during landing. The elevated leg design provides adequate ground clearance for onboard sensors while preventing contact with docking components. This frame–dock co-design approach enhances repeatability, safety, and positional accuracy during autonomous docking operations.

## 2. Intelligent Self-Cleaning Docking Station with Robotic Claws

The docking station functions as an intelligent autonomous unit for cleaning, alignment, and charging. Upon detecting a low-battery condition, the Ground Control Station triggers an ESP32-based controller to initiate a pre-docking cleaning sequence using servo-driven robotic claws with integrated brushes. During landing, the claws guide the drone's legs toward the copper charging plates to ensure proper alignment and retract after contact, enabling safe and reliable autonomous charging.

## 3. Polarity-Aware Contact-Based Charging Mechanism

The system employs an intelligent contact-based charging mechanism using two copper plates aligned with the drone's conductive landing legs. Upon docking, direct contact ensures efficient power transfer, while an automatic polarity check enables charging only after correct electrical alignment, preventing damage to onboard electronics. This simple and lightweight design avoids wireless charging losses and ensures reliable charging even under minor landing misalignments.